UNIVERSITI PENDIDIKAN SULTAN IDRIS

اونيۏرسيتي ڤنديديقن سلطان ادريس

SULTAN IDRIS EDUCATION UNIVERSITY

Software Verification and Validation (DES3103)

by

Dr. Nor Asiah Mohamad@Razak

Course Learning Outline

1. Briefing Session (Ice Breaking via Whiteboard, Class rules & Pro forma): **M. Teams**
2. Introduction to Software Verification & Validation I & II (V&V)
3. Managing V&V with SQA Integration
4. Quality metrics or attributes in V&V

WEEK 6-MID-TERM TEST (QUIZ): Topic 1 – 3-(CLO1:C2) – 20 MARKS

WEEK 7-PHASE 1: ASSIGNMENT 1-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO3:A5) - 20 MARKS: M. Teams

WEEK 9-PHASE 1: PRESENTATION & REFLECTION ON PHASE 1: M. Teams

5. Software Reviews – Will be covered on Assignments 1 & 2
6. Quality Control: Software Testing - Will be covered in Assignments 1 & 2
7. Human-Computer Interface Testing - Will be covered in Assignments 1 & 2
8. Manual Testing and Automation Testing: Will be covered in Final Assessment

WEEK 11-PHASE 2: ASSIGNMENT 2-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO2&4:P4&C5) - 20 MARKS: M. Teams
WEEK 13-PHASE 2: PRESENTATION & REFLECTION ON PHASE 2: M. Teams

9. Quality Maturity Models
10. Software Problem Analysis and Reporting
11. Quality Culture in doing V&V.
12. Safety and Security in V&V

WEEK 14: REFLECTION ON IMPLEMENTATION OPBL (SOFTWARE TESTERS) IN THE V&V COURSE: e-FORUM

WEEK 16-PHASE 3: FINAL ASSESSMENT – INDIVIDUAL SPECIAL PROJECT BASED ON PHASE 2 (CLO4:C5) (2 WEEKS) - 40 MARKS : e-ASSESSMENT

TOPIC 5

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Software Reviews

- Review objectives
- Formal design reviews (FDRs)
- Peer reviews (inspections & walkthroughs)
- Comparison between FDR and peer reviews methods
- Expert opinions

Software Reviews - Learning Outcomes

1. Review objectives
2. Formal design reviews (FDRs)
   - Characteristics of a leader
   - Participants
   - Preparations
   - The FDR session
   - Post-review activities

3. Peer reviews (inspections and walkthroughs)
   - Participants
   - Preparations
   - The FDR session
   - Post-review activities
   - Peer review coverage

4. Comparison between FDRs and peer reviews methods

5. Expert opinions

Software Reviews - Review Objectives

Four (4) direct objectives:

- To detect analysis and design errors as well as subjects where corrections, changes, and completions are required.
- To identify new risks likely to affect the project.
- To locate deviations from templates, style procedures and conventions.
- To approve the analysis or design product. The approval allows the team to continue on to the next development phase.

Two (2) indirect objectives:

1. To provide an informal meeting place for the exchange of professional knowledge about methods, tools, and techniques.
2. To record analysis and design errors that will serve as a basis for future corrective actions.

Software Reviews - Types of Formal Design Reviews (FDRs)

- DPR – Development Plan Review
- SRSR – Software Requirement Specification Review
- PDR – Preliminary Design Review
- DDR – Detailed Design Review
- DBDR – Data Base Design Review
- TPR – Test Plan Review
- STPR – Software Test Procedure Review
- VDR – Version Description Review
- OMR – Operator Manual Review
- SMR – Support Manual Review
- TRR – Test Readiness Review
- PRR – Product Release Review
- IPR – Installation Plan Review

Software Reviews - Characteristics of a FDR leader

01

* Knowledge and experience in development of projects of the type reviewed. Preliminary acquaintance with the current project is not necessary.

02

* Seniority at a level similar if not higher than that of the project leader.

03

* A good relationship with the project leader and his team.

04

* A position external the project team

Software Reviews - A FDR session

a. A short presentation of the design document.

b. Comments made by members of the review team.

c. Verification and validation of comments is discussed to determine the required action items (corrections, changes and additions).

d. Decisions about the design product (document), which determines the project's progress:

Full approval.	*Partial approval.*	*Denial of approval.*

Software Reviews - FDR Post-review activities

a. Preparation of the FDR report. The report's major sections:

- A summary of the review discussions.
- The decision about continuation of the project.
- A full list of the required action items — corrections, changes and additions. For each action item, completion date and project team member responsible are listed.
- The name(s) of the review team member(s) assigned to follow up.

b. Follow up performance of the corrections and to examine the corrected sections.

Software Reviews - Pressman's 13 golden guidelines for successful FDR

Design Review Infrastructure

1. Develop checklists for common types of design documents.
2. Train senior professionals serve as a reservoir for FDR teams.
3. Periodically analyze past FDR effectiveness.
4. Schedule the FDRs as part of the project plan.

The Design Review Team

5. Review team size should be limited, with 3–5 members being the optimum.

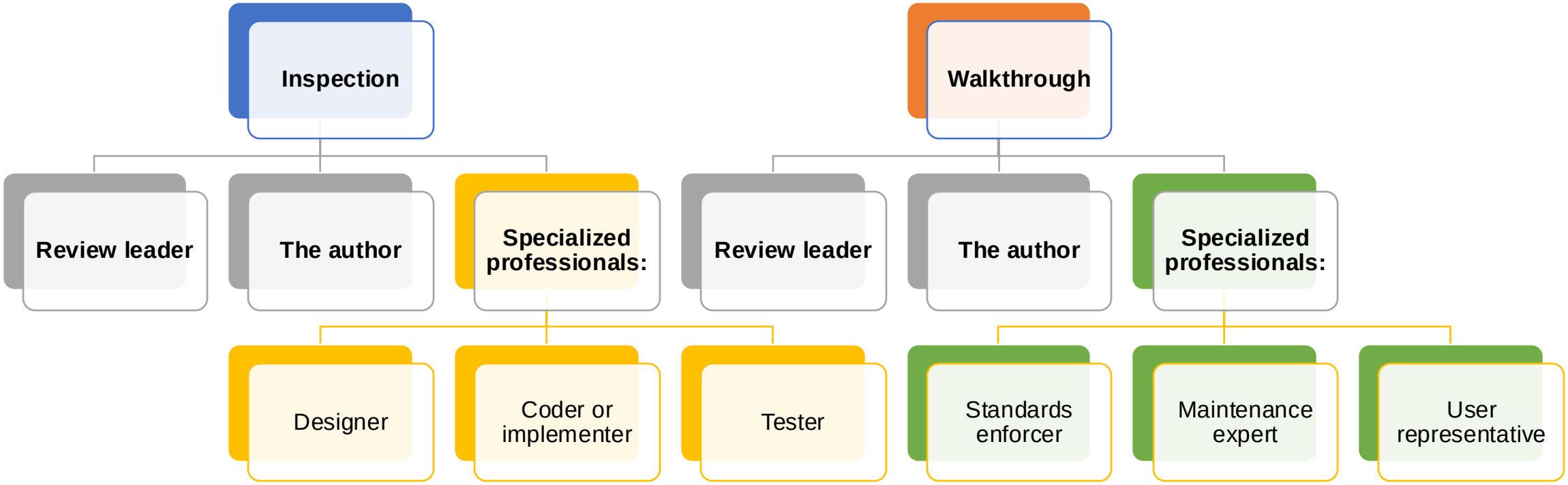
The Design Review Session

6. Discuss professional issues in a constructive way refraining from personalizing the issues.
7. Keep to the review agenda.
8. Focus on detection of defects by verifying and validating the participants' comments. Refrain from discussing possible solutions.
9. In cases of disagreement about an error - end the debate by noting the issue and shifting its discussion to another forum.
10. Properly document the discussed comments, and the results of their verification and validation.
11. The duration of a review session should not exceed two hours.

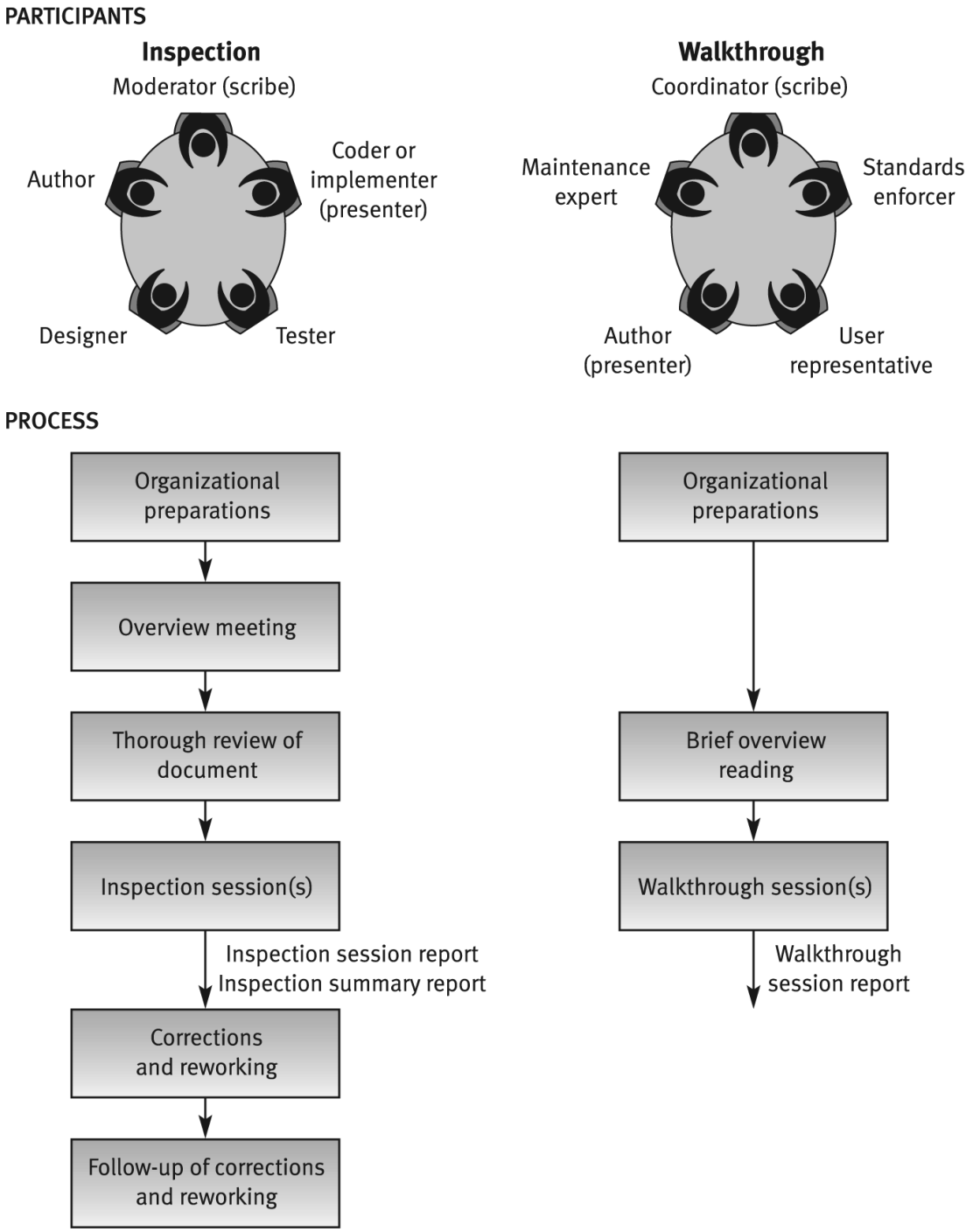
Post-Review Activities

12. Prepare the review report, including the action items.
13. Establish follow-up to ensure the satisfactory performance of all the list of action items.

Software Reviews - Participants of peer reviews



Software Reviews - Inspection VS. Walkthrough for participants' involvements



Software Reviews - Code inspection effectiveness at Fujitso (Cusumano)

Year	Defect detection method			Defects per 1000 lines of maintained code
	Test %	Design review %	Code inspection %	
1977	85	---	15	0.19
1978	80	5	15	0.13
1979	70	10	20	0.06
1980	60	15	25	0.05
1981	40	30	30	0.04
1982	30	40	30	0.02

UNIVERSITI PENDIDIKAN SULTAN IDRIS
SULTAN IDRIS EDUCATION UNIVERSITY

Software Reviews - Sections recommended to be included in or omitted from peer reviews

Sections recommended for inclusion

1. Sections of complicated logic.
2. Critical sections, where defects severely damage essential system capability.
3. Sections dealing with new environments.
4. Sections designed by new or inexperienced team members.

Sections recommended for omission

1. "Straightforward" sections (no complications).
2. Sections of a type already reviewed by the team in similar past projects.
3. Sections that, if faulty, are not expected to effect functionality.
5. Reused design and code.
6. Repeated parts of the design and code.

Software Reviews - Comparison of review methodologies – Process of review

Properties	Formal design review	Inspection	Walkthrough
Overview meeting	**No**	**Yes**	**No**
Participant's preparations	**Yes - thorough**	**Yes - thorough**	**Yes - brief**
Review session	**Yes**	**Yes**	**Yes**
Follow-up of corrections	**Yes**	**Yes**	**No**
Formal training of participants	**No**	**Yes**	**No**
Participant's use of checklists	**No**	**Yes**	**No**
Error-related data collection	Not formally required	Formally required	Not formally required
Review documentation	Formal design review report	**1)** Inspection session findings report **2)** Inspection session summary report	

Software Reviews - Why expert's opinion needed in reviews?

- Insufficient in-house professional capabilities in a specialized area.
- Temporary lack of in-house professionals for review team.
- Indecisiveness caused by major disagreements among the organization's senior professionals.
- In small organizations, where the number of suitable candidates for a review team is insufficient.

References

Galin, 2018, Software quality concepts and practice.

- Link : 2018, Daniel - Software quality _ concepts and practice- John Wiley & Sons.pdf

Galin, 2004, Software Quality Assurance- From Theory to Implementation.

- Link : 2004, Software Quality Assurance- From Theory to Implementation.pdf

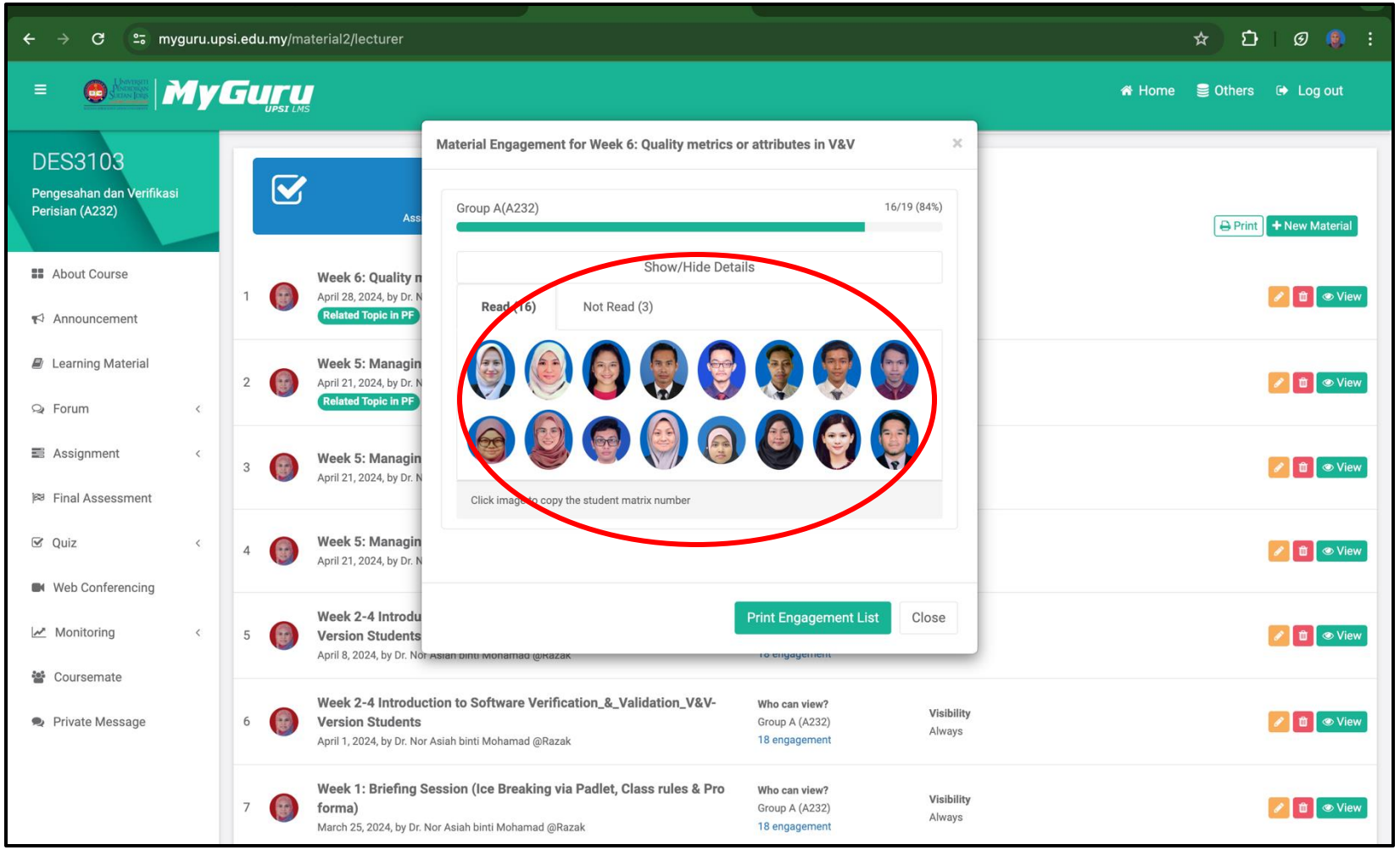
UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Activities…

Software Reviews - Task to do@MyGURU

For the purpose of attendance, please engage with the **“learning materials section”** as “shown in the red circle below” during class time.

Refer to the attached textbook, software quality assurance (Danial Gallin, 2004). Read on pages 149-171 of the textbook for answering the **Questions for class activity** in the next two (2) slides.



Software Reviews - Revision for final assessment (FA)

Questions :

1. Compare the various review techniques.

(1) In what aspects are design reviews more formal than inspections?

(2) In what aspects are inspections more formal than walkthroughs?

2. The chapter offers three different methodologies for team review of design documents.

(1) Which of the methodologies should a software development organization choose?

(2) Can more than one method be chosen and applied for the same document?

Alternatively, is it recommended to apply all three methods? List your arguments.

Software Reviews - Revision for final assessment (FA)

Questions :

3. The preparations made by members of inspection teams are considered to be of greater depth and thoroughness when compared with the preparations for walkthroughs.

(1) What activities are included in such high levels of preparation?

(2) Do you think that inspection teams having 15 members can achieve similarly high levels of preparation?

4. There are four direct objectives and two indirect objectives attached to the various review methods.

(1) List the direct and indirect objectives of each review method surveyed.

(2) For each objective, indicate the review technique or techniques that contribute(s) the most to achieving that objective.

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Additional notes…

Software Reviews - A sample general checklist for reviewing software documents

- Coverage and completeness
  - Are all essential items completed?
  - Have all irrelevant items been omitted?
  - Is the technical level of each topic addressed properly for this document?
  - Is there a clear statement of goals for this document?
  - (Don't forget: more documentation does not mean better documentation)
- Correctness
  - Are there incorrect items?
  - Are there any contradictions?
  - Are the any ambiguities?
- Clarity and Consistency
  - Are the material and statements in the document clear?
  - Are the examples clear, useful, relevant and correct?
- Clarity and Consistency (cont.)
  - Are the diagrams, graphs and illustrations clear, correct, use the proper notation, effective, in the proper place?
  - Is the terminology clear and correct?
  - Is there a glossary of technical terms that is complete and correct?
  - Is the writing style clear (nonambiguous)?
- References and Aids to Document Comprehension
  - Is there an abstract or introduction?
  - Is there a well placed table of contents?
  - Are the topics or items broken down in a manner that is easy to follow and is understandable?
  - Is there a bibliography that is clear, complete and correct?
  - Is there an index that is clear, complete and correct?
  - Is the page and figure numbering correct and consistent?

Adapted from Ilene Burnstein, Practical Software Testing, p. 327

Software Reviews - Typical tests and reviews

Reviews and testing

A software system is more than the code; it is a set of related artifacts; these may contain defects or problem areas that should be reworked or removed; quality-related attributes of these artifacts should be evaluated.

Reviews allow us to detect and eliminate errors/defects early in the software life cycle (even before any code is available for testing), where they are less costly to repair .

Most problems have their origin in requirements and design; requirements and design artifacts can be reviewed but not executed and tested.

Early prototyping is equally important to reveal problems in requirements and high-level architectural design.

A code review usually reveals directly the location of a bug, while testing requires a debugging step to locate the origin of a bug.

Adherence to coding standards cannot be checked by testing.

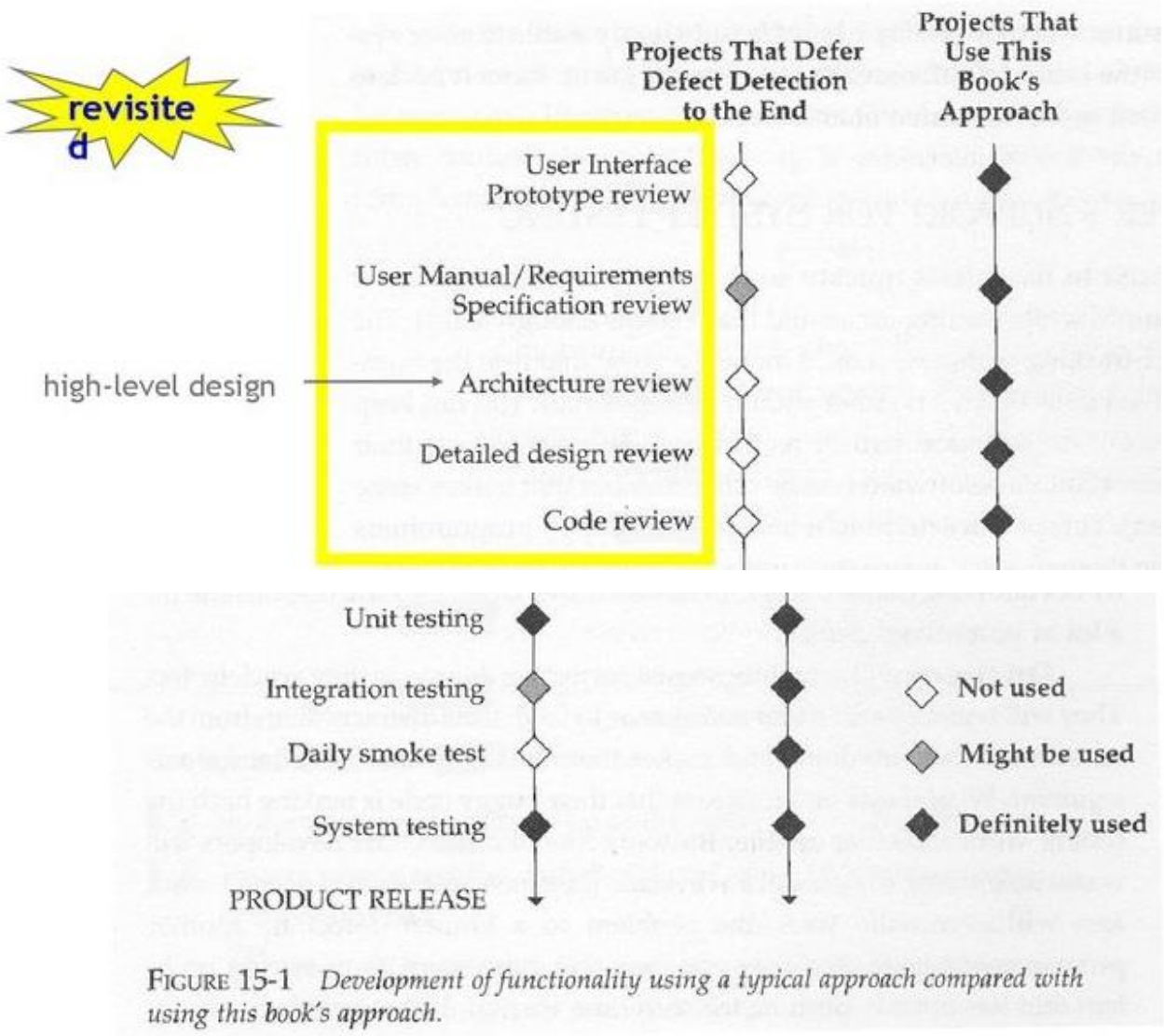


FIGURE 15-1 *Development of functionality using a typical approach compared with using this book's approach.*

Source: "Software Project Survival Guide", Steve McConnell

Software Reviews - Project documentation - *To build a successful product, all requirements should be written in detail and order – especially the **five main types**:*

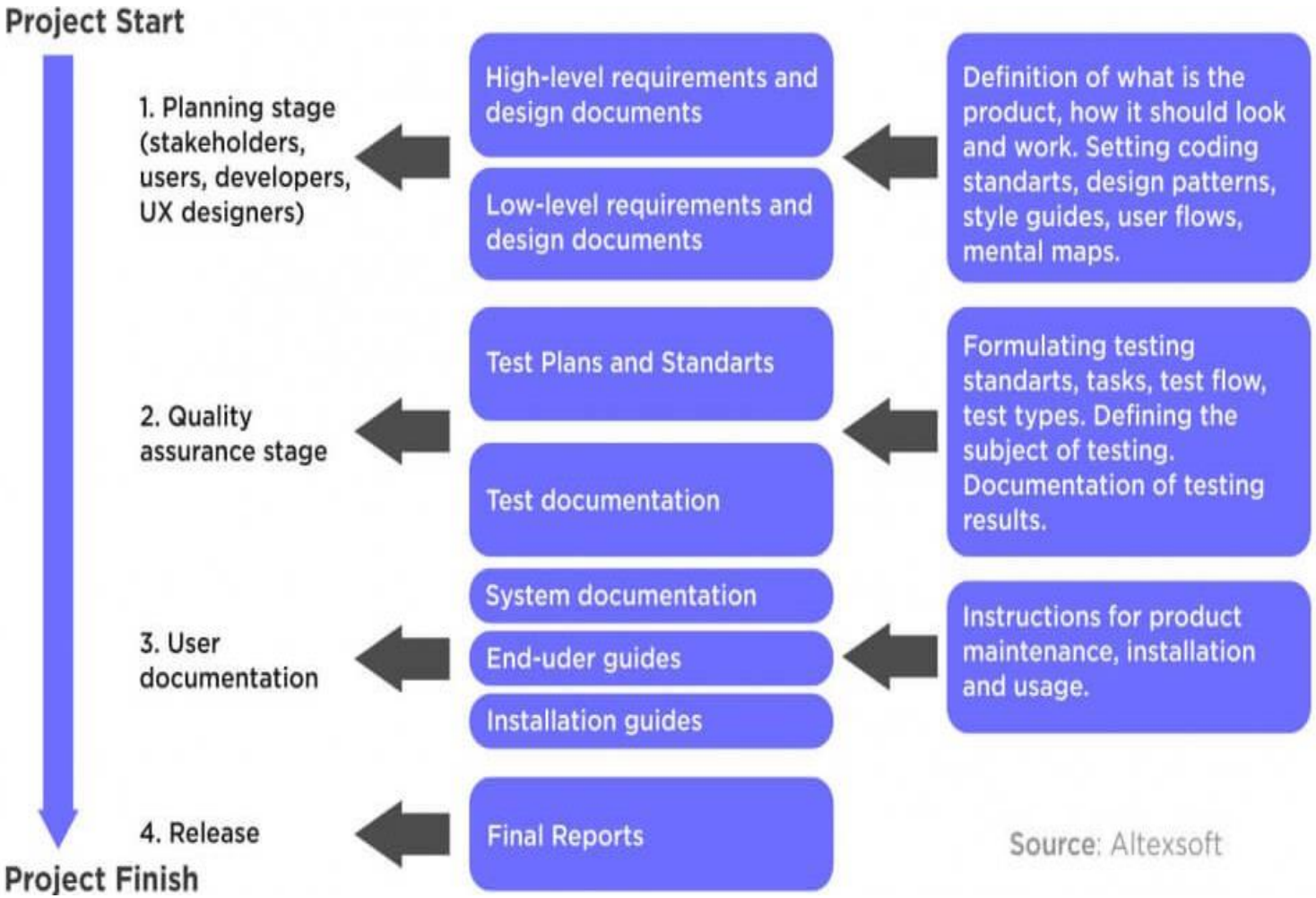
1.Business requirements: describe financial and marketing reasons that a company has for building the product. For instance, it can be increasing conversion by 20% or increasing request processing speed by 2 times.

2.User requirements: a development team analyzes the product from a user's point of view. The goal is to determine which functionality will improve user experience the most. An example of such requirements is "to create a customer profile that is easy to manage, and that can load fast even in low-bandwidth conditions."

3.Functional requirements: this type describes the features that should be implemented to achieve business and user objectives. For instance, it could be "to enable multi-factor authentication for increased profile security" or "to allow sorting product items by their price."

4.Non-functional requirements: these requirements describe rules that features have to comply with, but not the features themselves. For instance, a non-functional requirement for multi-factor authentication could be "to require a password with 8 numbers and 2 letters."

5. Implementation: requirements that describe technical ways of creating expected functionality – for instance, " to use JavaScript for a dynamic single-page application."



Software Reviews

Types of Reviews

END

UNIVERSITI PENDIDIKAN SULTAN IDRIS
اونيۏرسيتي ڤنديديقن سلطان ادريس
SULTAN IDRIS EDUCATION UNIVERSITY

Thank You…
Terima kasih..